



From the Beach to the Backyard: How Harmful Algal Blooms Impact Housing Renovation Decisions

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Abstract

Harmful algal blooms (HABs) impose a wide range of costs on surrounding communities. While the effects of HABs on real estate transactions are well studied, less is understood about whether homeowners make alterations to their homes in response to worsening water quality conditions. Using discrete-time duration analysis, 8 years' worth of satellite imagery, and residential construction permit data from six Ohio counties bordering Lake Erie, we find a positive and robust estimate of the elasticity of residential pool construction with respect to algal intensity. This suggests homeowners substitute a home renovation for recreational access. We later replace pool construction decisions with those for detached garages and find a negative relationship, indicating that alternative home renovations could be considered as complements to local environmental quality. These findings are consistent with the predictions derived from a theoretical model of homeowner utility maximization and they help to provide a more holistic understanding of the damages caused by HAB proliferation.

Keywords Algae · HAB · Housing · Renovation · Averting behavior · Control function

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1 Introduction

The localized presence of harmful algal blooms (HABs) has deleterious impacts on human health and regional economies (Kouakou and Poder 2019; Adams et al. 2018), while also impacting recreational decisions (Zhang and Sohngen 2018; Wolf et al. 2019). Despite a growing body of literature linking higher concentrations of HABs to lower housing values (Wolf and Klaiber 2017; Bechard 2021; Osseni et al. 2021), their full impact on household well-being is still relatively unknown, especially when considering decisions made throughout the duration of homeownership and not just at the time of sale.

Consider, for instance, households that decide to stay in areas plagued with HABs. Instead of relocating, these households might reduce their exposure by building a residential swimming pool in their backyard as an alternative to swimming at the beach. The influence of HABs on this type of decision isn't straightforward, owing to two competing effects. Firstly, just as air filters are used in response to heightened air pollution (Sun et al. 2017; Liu et al. 2018), homeowners might see residential pools as a substitute for degrading water quality in a nearby waterbody. Conversely, the presence of environmental hazards might lead homeowners to postpone or forgo home improvements either due to the prospect of relocating sooner than expected (Depro and Palmquist 2012), or because it might lower the returns on investments in a durable asset. Thus, the overall impact of these environmental issues on home renovation choices is ambiguous. An open and important question arises: when faced with deteriorating environmental amenities, do homeowners delay value-enhancing property investments?

We answer this question by estimating the influence of HABs on substitute (e.g., residential swimming pools) and complementary (e.g., secondary garages) renovation decisions using a capital-stock adjustment model (Helms 2003) and housing and remote-sensing algal data collected from six Ohio counties bordering Lake Erie between 2011 and 2018. Using a discrete-time duration model, we find a positive relationship between various measures of algal intensity and home renovations that substitute for the degraded environmental service. Our primary results suggest that pool construction has an elasticity of 4.04% with respect to nearby algal intensity, with the effect amplifying the closer a property is to Lake Erie. Conversely, we find that the construction of second garages—which are often constructed to store recreational equipment like motor boats, jet skis, or even kayaks and paddleboards—has an elasticity of -1.79% . Taken together, our findings indicate that households are more likely to make value-enhancing investments during periods of high HAB concentrations if the investment acts as a substitute for nearby water quality, and less likely if it serves as a complement. This aligns with predictions from our theoretical model.

Our findings are of both practical and academic importance. First and foremost, the results better reveal the full distributional impacts of HABs. The existing literature has primarily focused on decisions made by households relocating in response to poor water quality, whereas our analysis considers the costs borne by all homeowners, including those who choose to stay. Secondly, as HAB intensity is expected to worsen with climate change (Michalak et al. 2013), this analysis contributes to a body of literature studying the impact of climate change salience on residential construction decisions (McCoy and Zhao 2018; Barrage and Furst 2019). The paper proceeds with a description of the theoretical model and the empirical specification. We then briefly discuss the data before presenting the results. We conclude by considering the implications of our findings.

2 Theoretical Model

In our model, utility arises from housing services (h), environmental services (e), financial asset services (m), and consumption of other goods (z). Housing services are a function of initial building capital (k_0) and a renovation decision (r), given by $h(k_0, r)$.¹ Environmental services are a function of the neighborhood features (n) and the local presence of harmful algal blooms (HABs) (a), such that $e(n, a)$.² HAB presence impairs recreation through multiple channels, including changes in fishing activity (Wolf et al. 2017), boating decisions (Alvarez et al. 2019), and beach recreation (Zhang and Sohngen 2018; Wolf et al. 2019). Therefore, HABs are a detriment to environmental services ($\frac{\partial e}{\partial a} < 0$). Financial asset services are a function of the returns from the owned assets. Returns are impacted by algal blooms, neighborhood features, initial building capital, and the renovation decision, such that $m(a, n, k_0, r)$.³ There is evidence that HAB presence negatively capitalizes into property values ($\frac{\partial m}{\partial a} < 0$) (see Wolf and Klaiber 2017 or Bechard 2021).

This setup defines the individual problem to maximize utility through the choice of renovation (r) and consumption (z), given h , e , m , and the presence of a .

$$\text{Max}_{\{z, r\}} U = u(h(k_0, r), e(n, a), m(a, n, k_0, r), z) \quad (1)$$

With income (y), the household budget constraint is described by $y \geq z + p^r r$, where p^r is the price of renovation. Utility maximizing behavior across consumption and renovation produces first-order conditions (FOCs) that govern optimal decisions:

$$\frac{\partial \mathcal{L}}{\partial r} = \frac{\partial u}{\partial h} \frac{\partial h}{\partial r} + \frac{\partial u}{\partial m} \frac{\partial m}{\partial r} - \lambda p^r = 0 \quad (2)$$

To understand the potential impacts of HABs on renovation decision-making, we differentiate Eq. 2 with respect to algae (a). If neither the shadow price of the constraint (λ) nor the price of renovation (p^r) change with algae, the final term drops out (i.e., $\frac{\partial \lambda p^r}{\partial a} = 0$) and Eq. 3 is produced.⁴

$$\left[\frac{\partial^2 u}{\partial h \partial a} \left(\frac{\partial h}{\partial r} \right) + \frac{\partial^2 u}{\partial m \partial a} \left(\frac{\partial m}{\partial r} \right) + \frac{\partial^2 m}{\partial r \partial a} \left(\frac{\partial u}{\partial m} \right) \right] \quad (3)$$

The sign of Eq. 3 reveals algae's impact on the marginal benefit of renovation, which will influence the optimal renovation decision (r^*) that is obtained from maximizing Eq. 1 with respect to choice variable r .

¹ We assume $\frac{\partial u}{\partial h} > 0$, $\frac{\partial h}{\partial c} > 0$, $\frac{\partial c}{\partial r} > 0$.

² We assume $\frac{\partial u}{\partial e} > 0$ and $\frac{\partial e}{\partial n} > 0$.

³ We assume $\frac{\partial u}{\partial m} > 0$, $\frac{\partial m}{\partial k_0} > 0$, $\frac{\partial m}{\partial n} > 0$, and $\frac{\partial m}{\partial r} > 0$.

⁴ Housing services are also unimpacted by the presence of algae. It may be such that composite consumption is impacted by algal presence ($z(a)$). If this holds, a fourth term ($\frac{\partial U}{\partial z} \frac{\partial z}{\partial a}$) would be present in Eq. 3. The impact on optimal renovation is ambiguous since algal presence could increase consumption (e.g., more health-related expenditures) or reduce consumption (e.g., fewer purchases of recreational gear). We discuss the implication of this in greater detail in our concluding remarks.

Crucially, we must evaluate $\frac{\partial^2 u}{\partial m \partial a}$, $\frac{\partial^2 u}{\partial h \partial a}$, and $\frac{\partial^2 m}{\partial r \partial a}$ to determine the effect on renovation.

The term $\frac{\partial^2 u}{\partial m \partial a}$, which describes how HABs impact the marginal utility of the financial returns from renovation, is likely negative. The sign of the remaining terms will depend on the nature of the renovation-environment relationship. We will utilize three renovation cases to exposit the potential array of outcomes. Our first case involves a renovation that functions as a substitute for lake-based amenities, such as a private swimming pool. The marginal utility of housing services for this form of renovation increases as algae renders the public lake less enjoyable ($\frac{\partial^2 u}{\partial h \partial a} > 0$). The marginal impact of algae on financial returns ($\frac{\partial^2 m}{\partial r \partial a}$) for substitutable renovations is less clear. Prospective homebuyers may value a swimming pool as an offset to the loss of recreation potential, improving the financial returns of the renovation and the home with the presence of algae for existing homeowners. This leads to an ambiguous effect of algae on substitutable renovations like pool construction. Intuitively, a homeowner would be more interested in installing a pool when the lake is unavailable for recreation and there is the potential for additional renovation returns.

For the second case, suppose the housing service is provided through a renovation that complements the public good, such as installing a detached garage or a boathouse. These renovations may offer storage locations for watercraft devices or water-related activities (e.g., kayaks, boats, floats, etc.). In this example, the complementary effect strengthens the marginal utility of the housing service. If the lake is rendered unsuitable for recreation due to HABs, then the marginal utility from the housing services falls ($\frac{\partial^2 u}{\partial h \partial a} < 0$). Financial returns from complementary renovations will also decrease with the presence of algae since their value is also tied to the degrading environmental condition ($\frac{\partial^2 m}{\partial r \partial a} < 0$). Now, all three effects work together to reduce the likelihood of renovation decisions for renovations like secondary garages or boathouses (r^* decreases).

For the third case, consider a renovation decision unrelated to the environment's quality, such as exterior wall siding. The presence of algae does not strengthen nor diminish the marginal utility of the housing services provided by the renovation ($\frac{\partial^2 u}{\partial h \partial a} = 0$). Additionally, no renovation usage is tied to the environment, so there should be no financial related effects from the presence of algae ($\frac{\partial^2 m}{\partial r \partial a} = 0$). Here, only the utility impairment associated with financial returns from the property, given algae presence, are evaluated. Without offsetting or reinforcing impacts on utility, renovation decisions are less impacted than case 2, resulting in a reduced likelihood of installing exterior wall siding.

Our theoretical model suggests HABs can have mixed effects on renovation decisions. When considering renovation projects that substitute for environmental amenities, utility gains may outweigh the potential losses, leading to greater investment.⁵ Meanwhile, renovation projects unrelated to or complementary to environmental services should be less likely to occur, given the impairment in financial returns. Next, we develop an econometric model to evaluate our theoretical model's predictions, empirically.

⁵ It bears mentioning that while renovations such as residential swimming pools may be substitutes for the environmental public good, we do not consider the types of renovations considered in this paper to be substitutable for each other. Households that enjoy swimming in the lake are not necessarily households that enjoy boating/fishing in the lake and if a household becomes less interested in building a swimming pool, it does not necessarily cause them to become more interested in building a second garage.

3 Empirical Specification

The optimal renovation decision is achieved when the marginal rate of substitution between renovation and consumption is equated to the price of renovation. The theoretical setup above identifies the varying characteristics that impact the renovation choice, allowing us to cast this decision as.

$$r^* = r(n, k_0, a, p^r) \quad (4)$$

The household renovation decision depends on the level of r^* . If r^* is negative, then no expenditure would be made on renovations. If we let $\tilde{r}$ denote observable renovation expenditure, then

$$\tilde{r} = \begin{cases} 1 & \text{if } r^* > 0 \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

From Eq. 4, we observe that algae may impact the optimal renovation decision (r^*). Renovations that substitute for a degraded environment may be more likely to be built in the presence of HABs if utility gains outweigh financial losses ($\frac{\partial r^*}{\partial a} > 0$). This occurs through changes in renovation expenditures from $\tilde{r} = 0$ to $\tilde{r} = r^*$ if $r^*|(a=0) < 0$ and $r^*|(a \neq 0) > 0$. The exact opposite holds true if HAB presence diminishes the likelihood of renovation ($\frac{\partial r^*}{\partial a} < 0$). Assuming a linear form for Eq. 4, the optimal level of renovation for household i in year t is⁶

$$r_{it}^* = \beta^0 + \beta^1 n_{it} + \beta^2 k_{0it} + \beta^3 a_{it} + \epsilon_{it} \quad (6)$$

Taken together, Eqs. 5 and 6 imply that a household would renovate their home in year t ($\tilde{r}_{i,t}=1$) if their optimal level of renovation expenditures was positive and would not renovate their home in year t ($\tilde{r}_{i,t}=0$) if their optimal level of renovation expenditures was less than or equal to zero.

$$\tilde{r}_{it} = \begin{cases} 1 & \text{if } \beta^0 + \beta^1 n_{it} + \beta^2 k_{0it} + \beta^3 a_{it} + \epsilon_{it} > 0 \\ 0 & \text{if } \beta^0 + \beta^1 n_{it} + \beta^2 k_{0it} + \beta^3 a_{it} + \epsilon_{it} \leq 0 \end{cases} \quad (7)$$

While Eq. 7 can be estimated with a probit model, we maximize the efficiency of information available to us by considering the series of decisions that homeowners make across time. Indexing observations by year and household and including annual fixed effects yields a probit model that is statistically equivalent to a discrete-time duration model (Jenkins 1995; Beck 1998). The resulting data structure is pseudo-panel, with each observation cap-

⁶ Individual renovation prices are unobserved in our data. Given the small geographic extent of our analysis, renovation prices for comparable projects should be similar. We will capture changes in renovation prices over time through the inclusion of temporal fixed effects. However, differences in renovation size and scope, with different renovation prices, exist but are likely associated with observable housing characteristics (k_0). Therefore, the coefficient β^2 captures a mixture of preference and price influences on renovation.

turing the renovation decisions and algal values for a unique property and year combination. Consistent with the discrete-time duration modelling framework, each property is observed every year until “failure”, i.e., a renovation occurs ($\tilde{r}_{i,t}=1$). If $\tilde{r}_{i,t}=0$ for all years in the sample for a given property, then there are eight observations for that property. As our sample excludes properties with pre-existing focal renovations (pools in the case of our analysis of substitutable renovations and second garages in the case of complementary renovations), there aren’t any observations in our sample that have previously experienced a renovation. Our analysis therefore does not consider renovating behavior after a renovation has already been made.

4 Data

The data used in this analysis comes from three primary sources. County auditors from six Ohio counties bordering Lake Erie provided digital records of housing attributes. These records consisted of Computer Assisted Mass Appraisal (CAMA) files which include property characteristics (e.g., acreage, living area, structure age, etc.) used as control variables in our analysis, as well as Other Buildings and Yard Items (OBY) files which provided the date and address for residential pool construction permits.⁷ Our final dataset contains information for over 355,806 homes within 3,000 m of Lake Erie.⁸

HAB data is derived from 8-day composite satellite images of chlorophyll-a (CHLA) data, which is widely used as a proxy for HABs (Davis et al. 2012). These images are collected from NASA’s Ocean Color project. Although CHLA data is not consistently available for every 8-day period due to satellite interference, the data is generally extensive and available year-round. On average, each sampling location has 28 CHLA measurements taken per year, with over 1,200 sampling sites located within our study area. A snapshot of the resolution and spatial distribution of HAB data is illustrated alongside a map of our focal counties in Fig. 1.

In our baseline specification, we aggregate this data to the property-by-year level by first identifying the five closest sampling locations to each property and then averaging across all CHLA measurements taken at those sites. Since there are 28 measurements available per site per year, our property-by-year measure of HABs are derived from approximately 140 CHLA readings ($5 \times 28 = 140$). The bottom half of Fig. 1 shows an example area where the boundaries between each sampling location (i.e., raster cells) are clearly defined. Alternative CHLA aggregation methods are explored in the results section.

Because these satellite images are not exact measurements of water quality, they introduce measurement error, which may be a potential source of bias (Moore et al. 2020). To address this issue, we follow Wolf et al. (2022) by instrumenting for HABs using upstream nutrient loading measures from rivers that empty into Lake Erie. The approach used in this paper borrows the same logic but differs in two ways. First, instead of considering only loadings from the Maumee River, we expand the scope to include data from the next three

⁷ Because above ground pools do not typically require a permit, we are only able to comment on the impact of HABs on in-ground pool construction.

⁸ A spatial cutoff of 3,000 m was selected based on our review of the literature, which found that in large water bodies like the Chesapeake Bay (Walsh et al. 2017) or the Long Island Sound (Kung et al. 2022), water pollutants can influence housing market outcomes up to 2–3 km away.

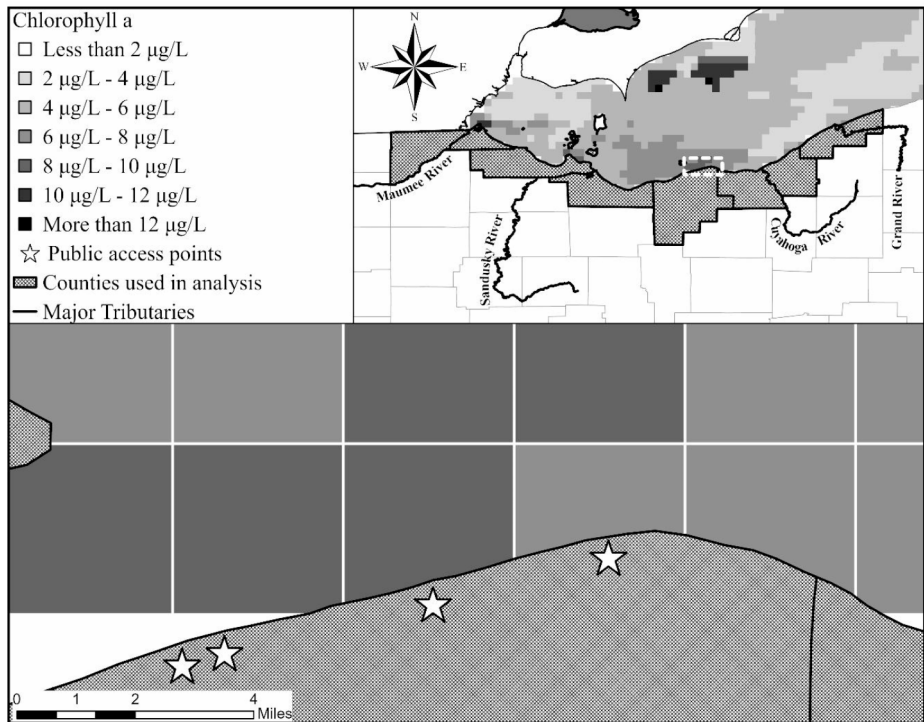


Fig. 1 Study Area. *Notes:* Local harmful algal bloom conditions are derived from lake-wide raster images of chlorophyll-a (CHLA) (see top half of Fig. 1). Each cell within the raster (white-bordered squares in the bottom half of Fig. 1) represents a distinct sampling location, with darker cells corresponding to higher CHLA. The average distance between each property and its five closest sampling sites is 3.8 km, with a standard deviation of 1.4 km. From west to east, the names of the counties used in this analysis are: Lucas, Ottawa, Erie, Lorain, Cuyahoga, and Lake

biggest tributaries, in terms of catchment area, in Ohio: the Grand River, the Cuyahoga River and the Sandusky River. This strategy strengthens our instruments but prevents us from using the same nutrient measure as Wolf et al. (2022), total phosphorus, as this data is not available for each river.⁹ Instead, we use the natural log of mean spring (March - June) discharge from the nearest of the four tributaries as an instrument. As greater levels of spring discharge should carry more phosphorous into the lake, which in turn promotes HAB formation in late summer/early fall (Michalak et al. 2013), we expect these instruments to be strong, and the first stage F-statistic reported in Tables 1 and 2 support this assumption. To provide even more information and overidentify our algal concentration measures, we also use spring discharge from the previous year as an instrument, as well as non-spring discharge from the current and previous year. Second, as our second stage has a nonlinear functional form, we use a control function approach rather than two-stage least squares (Georgic and Klaiber 2022; Wrenn et al. 2017).

⁹ As a robustness check, we mimic Wolf et al. (2022)'s approach by using total phosphorous as an instrument for HAB intensity in areas where total phosphorus data is available. Results from this alternative approach are similar in size and significance to those presented in the main text and are available upon request.

Table 1 Algal elasticity of pool construction

Variables	(1)	(2)	(3)	(4)
Algal Intensity	-0.073 (0.774)	4.043** (1.630)		
Algal Intensity within 500 m			1.638* (0.935)	5.463*** (1.667)
Algal Intensity Between 500 m and 1,000 m			0.204 (1.120)	3.942** (1.952)
Algal Intensity Between 1,000 m and 3,000 m			-1.361 (0.895)	2.396 (1.727)
Identification	Naïve	Control Function	Naïve	Control Function
First-Stage Test Statistic and [P-Value]		141.03 [<0.001]		141.03 [<0.001]
Over ID Test Statistic and [P-Value]		3.68 [0.299]		3.80 [0.284]
Observations	1,085,981	1,085,981	1,085,981	1,085,981
Pseudo R ²	0.146	0.147	0.146	0.146

***, **, * indicates significance at the 1%, 5% and 10% level, respectively. Reported coefficients represent elasticities and standard errors are calculated using the delta method and clustered at the Census block group level. Each model also includes property-specific characteristics (distance to the lake, distance to lake access, home age, home living area, home acreage, the number of bathrooms, the number of total rooms, and indicators for the presence of a fireplace, air-conditioning, a patio, and floodplain status), neighborhood characteristics (rainfall, temperature, land prices, average income, total additions, and variables capturing the percent of the neighborhood that is white, that is male, and that has a college degree), soil characteristics (slope and soil type), wind characteristics (speed, direction and their interaction), maximum Lake Erie water height, and year and county dummy variables.

Additional control variables include average rainfall and temperature measured at the Census block group level, whether the home is in a floodplain, indicators for soil type and the slope of the ground (which each help capture the difficulty of construction), wind speed and direction, and maximum Lake Erie water height. These variables come from the PRISM Climate Group, the Federal Emergency Management Agency, the Natural Resource Conservation Service, and the National Oceanic and Atmospheric Administration respectively.

5 Results

We begin by estimating Eq. 7 to quantify the impact of HABs on substitute renovation decisions. We incorporate neighborhood controls (n), such as average income and average annual precipitation, housing characteristics (k_0), such as the home age and size, and aggregated algal measures (a) with year and county fixed effects.¹⁰ The results, with stan-

¹⁰ Due to the relative infrequency of these housing renovations (only 0.02% of our property-year observations experience treatment, or about 1% of the homes in the sample), selecting any finer resolution for the spatial fixed effect results in a significant reduction in effective sample size due to loss in “within” variation.

Table 2 Algal elasticity of second garage construction

Variables	(1)	(2)	(3)	(4)
Algal Intensity	-0.397 (0.403)	-1.786* (1.057)		
Algal Intensity within 500 m			-0.712 (0.450)	-2.072** (1.059)
Algal Intensity Between 500 m and 1,000 m			-0.500 (0.527)	-1.895 (1.198)
Algal Intensity Between 1,000 m and 3,000 m			-0.155 (0.479)	-1.510 (1.088)
Identification	Naïve	Control Function	Naïve	Control Function
First-Stage Test Statistic and [P-Value]		151.45 [<0.001]		151.45 [<0.001]
Over ID Test Statistic and [P-Value]		5.13 [0.163]		4.58 [0.205]
Observations	1,041,794	1,041,794	1,041,794	1,041,794
Pseudo R ²	0.129	0.129	0.130	0.130

*** ** * indicates significance at the 1%, 5% and 10% level, respectively. Reported coefficients represent elasticities and standard errors are calculated using the delta method and clustered at the Census block group level. Each model also includes all additional control variables listed under Table 1.

dard errors clustered at the Census block group level are displayed in Table 1. The first two columns present estimates of the elasticities of pool construction in response to increasing algal concentrations. The coefficients reported in column (1) are from a naïve specification in which algal concentrations are related to pool construction within a probit model, while the coefficients from a control function/instrumental variable (IV) approach (Georgic and Klaiber 2022; Wrenn et al. 2017) are shown in column (2). Two conclusions are readily apparent. First, the estimate produced by the control function approach is positive and statistically significant, suggesting that a 1% increase in nearby average algal concentrations is associated with a 4.04% increase in the likelihood of building a pool. Because this estimate is positive, it indicates that the positive marginal utility gains from renovation dominate the negative and potentially ambiguous terms related to the financial returns from renovation that are present in Eq. 3.¹¹ This suggests that homeowners view residential swimming pools as substitutes for recreational opportunities provided by Lake Erie, opting for pools as lake water quality declines. A second key takeaway from Table 1 is that the naïve estimate is statistically insignificant and over an order of magnitude smaller, which is consistent with prior studies (Keiser 2019). The first stage results for the model in column (2) are reported in Appendix Table 1.

Columns (3) and (4) display a similar attenuation bias in the naïve estimates. However, the models in columns (3) and (4) differ from those in columns (1) and (2) by allowing the impact of algae on renovation decisions to vary spatially. This is achieved by including spatial dummies indicating whether a property is located 0–500 m, 500–1000 m, or 1000–3000

For example, replacing county fixed effects with Census tract fixed effects eliminates half of the observations. We also estimated a county-by-year fixed effects model which continued to show a positive and statistically significant relationship between HAB intensity and pool renovation decisions. Results from this alternative specification are available upon request.

¹¹ While separately identifying the individual terms of this equation would be interesting and informative, the estimates reported in this paper describe the net effects of changes in algal conditions on renovation decisions.

m from Lake Erie and then interacting these terms with the algal measure. We note that due to perfect collinearity both the level algal term and the 1000–3000 m spatial dummy are dropped from the model. Column (4) intuitively demonstrates that the treatment effect is strongest for homes closest to the lake, gradually diminishing with distance and becoming statistically insignificant for homes located more than 1,000 m away. Lastly, overidentification test statistics reported in columns (2) and (4) of Tables 1 and 2 suggest the IVs are valid and properly excluded from the second-stage. These tests, in other words, suggest river discharge is only related with housing renovation decisions through their impact on algal intensity.

Next, we assessed the impact of HABs on secondary garage installations by re-estimating each of the models from Table 1, but using the construction of secondary garages as the dependent variable rather than the construction of swimming pools. Overall, the negative coefficients from Table 2 suggest that homeowners view second garages as a complement to water quality, which aligns with predictions from our theoretical model. In this case, a 1% increase in nearby algal concentrations is associated with a 1.79% decrease in the likelihood of building a second garage. Table 2 indicates that homeowners are more likely to invest in value-enhancing renovations when water conditions improve, particularly those that help them better utilize the lake. For example, second garages are often used to store recreational vehicles like boats or jet skis, which can make lake trips more enjoyable. Shifting focus to column (4), a similar attenuation bias to that seen in Table 1 is evident. Specifically, the magnitude of the estimated impact of water quality on second garage renovations diminishes across the spatial gradient, with the strongest effects observed for homes closest to the lake.

5.1 Testing Underlying Assumptions of the Instrumental Variable Approach

Instrumental variables can address measurement error if the applied instruments are relevant, exogenous, and satisfy monotonicity (Cunningham 2021). The first-stage F-values from Tables 1 and 2 provide strong evidence of the first condition, indicating a high correlation between the instruments and the remote-sensing measure of HABs. As a general rule of thumb, instruments are considered to be relevant and not subject to weak instrument issues when the first-stage F-value exceeds 10 (Stock and Watson 2020; Cunningham 2021). This threshold is easily surpassed across all specifications.

Regarding the second assumption—that the instruments are uncorrelated with the error term—the overidentification tests in Tables 1 and 2 provide some, but not definitive, evidence of this assumption being met. In each model, we fail to reject the null hypothesis that discharge is unrelated to renovation decisions other than through its impact on algae. However, failing to reject the null does not guarantee instrument exogeneity as overidentification tests rely on residuals being an accurate approximation of the true error values, which are inherently unobservable.

Given the limitations of the overidentification test, it is important to provide more evidence of instrument exogeneity. One potential confounder is flooding. River discharge could directly influence homeowner recreation and renovation decisions by flooding nearby

access points, making construction of private pools more appealing. If such a relation exists, then our instruments would violate the exclusion restriction, as discharge should be directly included in the second-stage IV regression.

However, we find no evidence supporting this concern. First, we include an annual maximum measure of Lake Erie water height within the first and second-stage regression. This controls for any large shifts in water levels across time and space. Second, while the four tributaries in the study area contribute a notable share of nutrients to the lake, their discharge rates are relatively minor compared to Lake Erie's largest tributary, the Detroit River, which accounts for 80% of the lake's water (Scavia et al. 2023). Thus, significant increases in discharge from any Ohio tributary are unlikely to have a meaningful impact on lake-wide water levels. Last, when plotting monthly discharge rates against water heights using data from the mouth of the Maumee River, the largest tributary in the study area, we find no evidence of a strong correlation between the two (Appendix Fig. 1).

Rainfall could also be another possible confounder as it could be correlated with discharge and a homeowner's decision to build an outdoor pool (i.e., homeowners may see less value in a pool if it is raining frequently). Similar to the case above, we directly control for this by including rainfall, measured at the Census block group by year level, as an independent variable within the first and second-stage regression. Other weather-related controls are also added, including wind speed, wind direction and the interaction between the two.

We now examine whether discharge satisfies the monotonicity condition. In this context, monotonicity requires that discharge increases HAB exposure for all homeowners rather than increasing exposure for some while reducing it for others. To test this, we first assess whether discharge from each river positively influences nearby HAB growth. For monotonicity to hold, discharge from each river should, at a minimum, have a positive effect on HABs.

We begin by creating four river dummy variables, each identifying the river closest to a given house. Discharge is then interacted with these dummies and included as predictors of algal intensity, along with all of the original first-stage controls. The results from this modified first-stage specification are reported in Appendix Table 2. Moving from left to right, we see that both spring discharge (column 1) and non-spring discharge (column 2) positively affect HAB intensity, regardless of which river is considered. Monotonicity also requires that discharge positively influence HAB growth regardless of the period considered. Similar to columns 1 and 2, we investigate this concern using interaction terms. Discharge is interacted with four time period dummies, with each period composed of two years (2011–2012, 2013–2014, etc.), and included as regressors in columns 3 and 4. Overall, we do not find evidence that discharge is positively influencing algal intensity in some periods but negatively influencing it in others, suggesting that monotonicity is upheld.

5.2 Sensitivity analysis

While we would expect homeowners to perceive changes in algal concentration, it is not objectively clear how to model this perception. We have presumed that homeowners make renovation decisions in response to continuous measures of algal intensity in the present

Table 3 Algal measurement robustness

Variables	(1)	(2)	(3)	(4)	(5)	(6)
Algal Intensity	-0.242 (0.609)	4.326** (1.988)	-0.193 (0.947)	4.057*** (1.473)	-0.057 (0.389)	3.434** (1.521)
Identification	Naïve	Control Function	Naïve	Control Function	Naïve	Control Function
Number of Sampling Locations Considered	3	3	7	7	5	5
Average or Maximum Algal Concentrations	Average	Average	Average	Average	Maximum	Maximum
First-Stage Test		52.50		168.95		41.55
Statistic and [P-Value]		[<0.001]		[<0.001]		[<0.001]
Observations	1,085,981	1,085,981	1,085,981	1,085,981	1,085,981	1,085,981
Pseudo R ²	0.145	0.149	0.145	0.147	0.145	0.146

***, **, * indicates significance at the 1%, 5% and 10% level, respectively. Reported coefficients represent elasticities and standard errors are calculated using the delta method and clustered at the Census block group level. Each model also includes all additional control variables listed under Table 1.

year averaged across the five nearest measurements. However, homeowner perceptions could be formed from a more narrow or expansive set of observations. Similarly, though our discharge instruments appear to be valid, different sets of valid instruments may lead to qualitatively different results. We assess the sensitivity of our results to such modeling decisions in Tables 3 and 4, and 5.

Though it is challenging to consider every permutation, as the models in Tables 3 and 4 suggest, homeowners have a qualitatively similar response regardless of how nearby HAB conditions are measured. Models (1) and (2) in Table 3 restrict the number of nearby algal measurements to come from the three nearest locations rather than the closest five. Alternatively, Models (3) and (4) expand the number from five to seven. Both sets of models display qualitatively similar results to those presented in Table 1; there is a positive and statistically significant relationship between algae and pool construction when considering estimates from the control function approach. A similar pattern is revealed in models (5) and (6) when replacing *average* algae from the five nearest locations in a given year with the *maximum* reading in that year.

Since homeowners may be influenced by both current and historic water quality conditions, it is worthwhile to test how sensitive the results are to different temporal definitions of water quality. There are two relevant time dimensions in our analysis; when HAB intensities are measured and when river discharge volumes are recorded. Column (1) of Table 4 presents results from the primary specification (model (2) from Table 1), which uses discharge values from the present and previous year to instrument for present-year algae. Column (2) considers a deeper instrumental lag, using discharge volumes from the previous year as well as the year preceding it. This approach eliminates concerns about simultaneous effects of weather on both discharge and renovation decisions that could potentially endogenize the instruments. The similarity between the estimates reported in columns (1) and (2) suggest this is not likely a concern.

Table 4 Algal and instrument lag robustness

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Current Year Algal Intensity	4.043** (1.630)	4.514** (1.874)	8.636*** (3.234)	6.545*** (2.293)	3.887** (1.572)	4.113* (2.455)	6.647*** (2.329)	2.026 (1.709)
Lagged Algal Intensity			4.924 (3.292)	2.274 (2.062)	7.284 (5.313)	-8.542 (10.677)	8.986 (6.501)	9.261 (6.480)
Algal Lags	Zero Years	Zero Years	Previous Year	Previous Year	Average of Two Preceding Years	Average of Two Preced- ing Years	Average of Three Preced- ing Years	Average of Three Preced- ing Years
Standard Instruments (Discharge in Current and Previous Year)	Yes		Yes		Yes		Yes	
Lagged Instruments (Discharge One Year Prior to Cur- rent and Lagged Algal Measure)		Yes		Yes		Yes		Yes
Current Year Algal Intensity First- Stage Test Statistic and [P-Value]	141.03[<0.001]	83.42[<0.001]	144.63[<0.001]	90.32[<0.001]	144.63[<0.001]	85.62[<0.001]	144.63[<0.001]	69.83[<0.001]
Lagged Algal Intensity First-Stage Test Statistic and [P-Value]			122.04[<0.001]	50.56[<0.001]	31.27[<0.001]	23.10[<0.001]	138.90[<0.001]	27.93[<0.001]
Observations	1,085,981	1,085,981	1,085,981	1,085,981	1,085,981	1,085,981	1,085,981	1,085,981
AIC	3710.095	3708.765	3709.834	3715.893	3710.985	3718.398	3712.448	3719.17
BIC	4221.709	4196.583	4233.345	4251.303	4234.497	4241.91	4247.858	4242.682
Pseudo R ²	0.147	0.147	0.148	0.148	0.148	0.146	0.148	0.146

***, **, * indicates significance at the 1%, 5% and 10% level, respectively. Reported coefficients represent elasticities and standard errors are calculated using the delta method and clustered at the Census block group level. Each model also includes all additional control variables listed under Table 1.

Columns (3) through (8) of Table 4 introduce lagged measures of algal intensity alongside the current-year measure included in the primary model. Columns (3) and (4) incorporate variables for algal intensity from both the current and previous year. Columns (5) and (6) replace the previous year's algal intensity with the average intensity from the past two years, while columns (7) and (8) use average algal intensity from the past three years. The odd-numbered columns use the standard set of instruments used throughout the rest of the paper: spring and non-spring discharge for the nearest tributary in the present year and in the previous year. The even-numbered columns use lagged instruments, including the discharge values for the previous year as well as for the year preceding the deepest lag. Interestingly, the lagged value isn't a significant determinant of pool construction in any permutation. While this may seem surprising at first, it is consistent with several valuation studies which find individuals are more responsive to recent rather than historic measures of environmental quality (Wolf and Klaiber 2017, Chen et al. 2019; Mamun et al. 2023).

Lastly, we consider the sensitivity of our estimates to a number of modelling decisions in Table 5. Column (1) uses new instruments that are produced by interacting our standard instruments with the distance from each property to the nearest tributary to assuage concerns that our instruments better predict algal levels for homes closer to the tributary than for those further away. The similarity between the estimate reported in column (1) of Table 5 and that reported in column (2) of Table 1 give us some confidence that this is not a significant concern. Because homebuyers may be less aware of local algal conditions than existing homeowners, and may thus be less likely to invest in substitutable renovations, we consider the impact of omitting two types of observations related to property transactions in columns (2) and (3) of Table 5. Column (2) omits all observations for properties which were sold over the course of our sampling timeframe. Column (3) only omits observations for years in which it was sold. The estimates reported in both columns are qualitatively similar to our primary results. Since the estimates are slightly greater in magnitude, this suggests new homebuyers may be less likely to renovate in response to HABs relative to existing homeowners. Additional research on this topic is needed, however, as keeping only non-transacted properties in our analysis could lead to sample selection bias. Finally, columns (4) and (5) consider alternative clustering levels for the standard errors, yielding results of similar significance to our primary estimates.

6 Discussion and Conclusion

Using satellite imagery and eight years of home renovation records across six counties in northern Ohio, we estimate a series of probabilistic regressions that provide strong evidence of nearby environmental quality influencing homeowner renovation decisions. Using residential pool construction as an example, we find homeowners are more likely to construct a pool when nearby water conditions on Lake Erie worsen due to the presence of HABs. This suggests homeowners are able to offset some of their losses from a nearby pollutant through private construction, so long as the renovation and environmental good are, to some degree, substitutable.

Table 5 Modeling decision sensitivity analysis

Variables	(1)	(2)	(3)	(4)	(5)
Algal Intensity	3.529**	6.144***	4.811***	4.043**	4.043***
	(1.380)	(2.014)	(1.712)	(1.871)	(0.909)
Instruments	River Dis-charge Interacted with Dis- tance to Nearest Tributary	River Dis-charge	River Discharge	River Discharge	River Discharge
Transactions Excluded	None	Yes if Property Sold At Least Once Between 2011 and 2018	Yes for All Years In Which Proper-ties Sold	None	None
Clustered Standard Errors	Block Group	Block Group	Block Group	Tract	County
First-Stage Test Statistic and [P-Value]	137.93 [<0.001]	113.79 [<0.001]	135.91 [<0.001]	50.4 [<0.001]	226.28 [<0.001]
Observations	1,085,981	633,035	1,004,598	1,085,981	1,085,981
Pseudo R ²	0.146	0.161	0.150	0.145	0.146

***, **, * indicates significance at the 1%, 5% and 10% level, respectively. Reported coefficients represent elasticities and standard errors are calculated using the delta method and clustered at the Census block group level. Each model also includes all additional control variables listed under Table 1.

This substitution behavior is more strongly observed the closer a property is to the lake and is robust to a variety of modeling decisions. In our primary specification, we instrument for current algae conditions using current and past river discharge measures from the four largest tributaries in Ohio, while also providing empirical evidence that our instruments are exogenous, relevant and satisfy monotonicity. Using the same methodology, we also find that homeowners delay the construction of detached garages in response to elevated algal intensity, suggesting that environmental quality's effect on home renovation decisions depends on whether the home renovation acts as a substitute or complement to the local environmental service.

Our findings indicate that both swimming pool and detached garage construction are elastic in response to water quality and algal intensity, though pool construction appears to be more responsive to HABs. As a point of reference, the probability a homeowner in our sample builds a pool within a given year is 0.02%. Our primary model suggests that if average chlorophyll-a concentrations in our study area (6.815 ug/L) doubled, rendering

water quality conditions unsafe for recreation in many parts of the study area, the incidence of swimming pool construction would increase to only roughly 0.10% per year. Though the annual likelihood of the average home adding a pool is still low in absolute terms, the relative change under these hypothetical worsened conditions is considerable with the average homeowner being five times more likely to build a pool in a given year ($5 = 0.10\%/0.02\%$) relative to current conditions.

While the presence of HABs is known to impair housing prices and reduce fishing permit revenue (Wolf and Klaiber 2017; Wolf et al. 2017), our results suggest HABs impose additional, more opaque, costs within a community. As homeowners respond to deteriorating lake quality through increased residential swimming pool construction, revenue is generated for local businesses. As this revenue is subject to a regional spending multiplier, there is an argument that this substitution effect is an economic stimulus. This would, however, be a misguided argument and is tantamount to the “Broken Window Fallacy” (Bastiat 1850). The elevated algal intensity does not create wealth or value for these communities, and the additional money spent on swimming pools could have been spent elsewhere. Ultimately, “society loses the value of the objects unnecessarily destroyed”, in this case, the recreational value of going to the beach.

Our findings highlight a shift in homeowner behavior and underscore a broader economic impact: increased local expenditures in response to environmental degradation should not be mistaken for a positive economic outcome. Such spending is reactive and compensatory, indicating not gains but a redistribution of community resources. Furthermore, pool construction costs are large (between \$20,000 and \$50,000) and require significant disposable income.¹² We show that residential pools are a form of renovation that functions as a substitute for a degraded environment. However, the ability to mitigate utility loss from degraded environments through a durable investment will be constrained to homes with sufficient disposable income. This constraint may preclude homes with lower socioeconomic status from investing in mitigating adaptations, leading to greater utility impairment. This inequity suggests that the true cost of HABs extends beyond diminished property values and lost recreation opportunities. Future research utilizing household expenditure data could further identify some of these more complex welfare effects and better inform our collective understanding of the distribution of costs of HABs on Ohio’s Erie coast.

Appendix

See Tables 6, 7 and Fig. 2

¹² National Association of Realtors. 2010. <https://www.nar.realtor/blogs/styled-staged-sold/are-pools-worth-the-expense>.

Table 6 First stage results

Variables	(1)
<i>Excluded Instruments</i>	
Spring Discharge [†]	1.493*** (0.211)
Spring Discharge for Previous Year [†]	-0.454*** (0.013)
Non-Spring Discharge [†]	1.500*** (0.150)
Non-Spring Discharge for Previous Year [†]	-0.526*** (0.083)
<i>Block Group Climatic Variables</i>	
Precipitation [†]	-2.130*** (0.230)
Temperature [†]	-0.661 (0.828)
Wind Speed [†]	-145.520*** (18.183)
Wind Direction [†]	-46.478*** (6.244)
Wind Direction X Speed [†]	27.559*** (3.444)
High Water [†]	-110.214*** (38.817)
<i>Block Group Socioeconomic Characteristics</i>	
Proportion Male	-0.461* (0.274)
Proportion with College Degree	-0.455** (0.214)
Proportion White	0.825*** (0.108)
Average Income [†]	-0.029 (0.047)
Land Price [†]	0.111** (0.056)
Properties with Additions [†]	0.070*** (0.024)
<i>Property Characteristics</i>	
Acreage [†]	-0.035* (0.018)
Distance to Lake [†]	0.094*** (0.020)
Distance to Access [†]	0.033 (0.028)
Floodplain (0/1)	-0.127** (0.055)
High Slope (0/1)	-0.332*** (0.097)
Soil Type 1 (0/1)	0.019 (0.084)
Soil Type 2 (0/1)	0.042 (0.092)
Soil Type 3 (0/1)	-0.060 (0.081)
<i>Home Characteristics</i>	
Existing Patio/Deck (0/1)	-0.028*** (0.007)
Home Age [†]	-0.007 (0.023)
Home Square Feet [†]	0.079*** (0.023)
Air Conditioning (0/1)	-0.051*** (0.009)
Fireplace (0/1)	0.012 (0.010)
Number of Rooms	0.006* (0.004)
Number of Bathrooms	0.020** (0.008)
Observations	1,085,981
R ²	0.8229

***, **, * indicates significance at the 1%, 5% and 10% level, respectively. [†] represents a logged variable. Model also includes county and year fixed effects.

Table 7 Instrument monotonicity

Dependent Variable: Algal Intensity	(1)	(2)	(3)	(4)
	Spring Discharge	All Other Discharge	Spring Discharge	All Other Discharge
Ln Discharge * Cuyahoga River (0/1)	1.620*** (0.211)	1.940*** (0.097)	-	-
Ln Discharge * Grand River (0/1)	1.529*** (0.211)	1.866*** (0.099)	-	-
Ln Discharge * Maumee River (0/1)	1.660*** (0.193)	1.968*** (0.089)	-	-
Ln Discharge * Sandusky River (0/1)	1.709*** (0.207)	1.995*** (0.094)	-	-
Ln Discharge * Year11or12 (0/1)	-	-	1.689*** (0.209)	1.757*** (0.108)
Ln Discharge * Year13or14 (0/1)	-	-	2.235*** (0.201)	2.509*** (0.101)
Ln Discharge * Year15or16 (0/1)	-	-	1.333*** (0.165)	1.526*** (0.095)
Ln Discharge * Year17or18 (0/1)	-	-	2.448*** (0.176)	2.223*** (0.111)
Observations	1,085,981	1,085,981	1,085,981	1,085,981
R ²	0.806	0.819	0.813	0.823

Notes: ***, **, * indicates significance at the 1%, 5% and 10% level, respectively. Each model includes all additional control variables listed under Table 1. Standard errors are clustered at the Census block group level. Algal Intensity is a measure of mean chlorophyll-a from the 5 closest monitoring stations. Each regression relates discharge with algal intensity using measures taken from the same year.

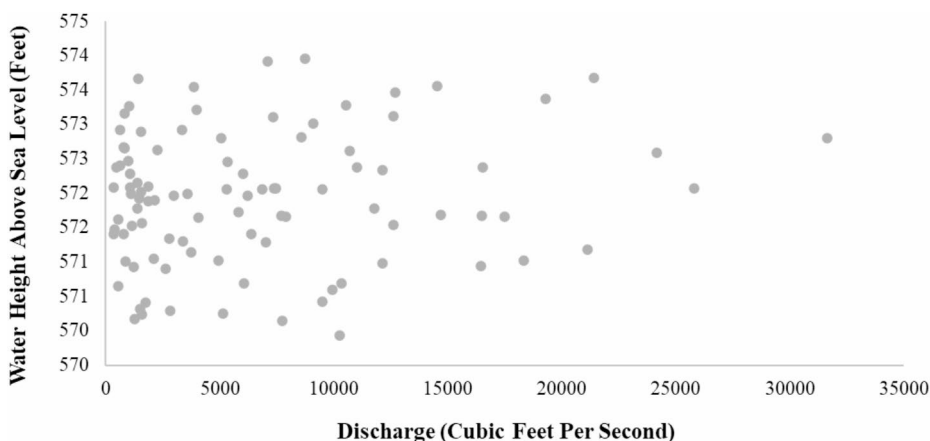


Fig. 2 Lake Erie water height vs Maumee River discharge. *Note:* Mean monthly water height (y-axis) is plotted against monthly discharge rates from the Maumee River (x-axis). Data was collected from the USGS, station number 4193500, between 2011 and 2018. A simple regression between the two does not reveal a statistically significant relationship ($p\text{-value} = 0.16$)

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